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Environmental covariates of sablefish (*Anoplopoma fimbria*) and Pacific ocean perch (*Sebastes alutus*) recruitment in the Gulf of Alaska



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ARTICLE INFO

Available online 25 February 2015

Keywords:

Structural analysis
Recruitment
Environmental effects
Larval settlement
Horizontal advection

ABSTRACT

The sablefish (*Anoplopoma fimbria*) and Pacific ocean perch (POP; *Sebastes alutus*) fisheries in the Gulf of Alaska (GOA) are both highly lucrative and variable. Determining environmental factors that drive variability in their recruitment may improve our understanding of forces affecting their early life survival, which may be helpful when evaluating management strategies. Here we examine relationships between sablefish and POP recruitment and multiple environmental indices associated with circulation in the GOA. We used hierarchical cluster analysis to determine spatially and seasonally relevant scales for analyzing these relationships. We then used structural equation modeling to examine sequential relationships between large-scale climate variables, regional (eastern and western GOA) environmental variables, and recruitment using both hypothesis-testing and exploratory approaches. Exploratory analyses revealed that sablefish recruitment was positively related to July upwelling-favorable winds and negatively related to late winter freshwater discharge in the eastern GOA during age 1. POP recruitment was negatively related to June upwelling-favorable winds in both regions during ages 0 and 1 and positively related to late spring freshwater discharge throughout the GOA during age 1. These results suggest that upwelling-favorable winds and freshwater discharge may affect recruitment of both species through productivity-related mechanisms, and may additionally affect POP recruitment through advection-related mechanisms. Targeted studies at the appropriate scales are needed to provide greater certainty in the potential mechanisms behind these relationships.

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1. Introduction

Sablefish (*Anoplopoma fimbria*) and Pacific ocean perch (POP, *Sebastes alutus*) both support highly lucrative commercial fisheries in the Gulf of Alaska (GOA). Identifying environmental drivers of recruitment may lead to a greater understanding of mechanistic processes affecting early life survival of these species (e.g., McFarlane and Beamish, 1992; Vestfals et al., 2014; Shotwell et al., 2014). Furthermore, robust relationships between recruitment and environmental variables, if well understood, may be incorporated into simulations used for management strategy evaluations (MSEs) in order to inform potential management measures under varying environmental conditions (e.g., A'Mar et al., 2009; Ianelli et al., 2011). This study seeks to determine the biological relevance of several specific climate and regional-scale environmental processes (as revealed by derived indices) to sablefish and POP recruitment in the GOA, as well as to identify

the specific regions within the GOA that exhibit the strongest recruitment–environment relationships.

1.1. Sablefish

Sablefish are a deep-sea fish inhabiting waters throughout the North Pacific, with a southern range limit extending from Sagami Bay, Japan in the western Pacific to Baja California, Mexico in the eastern Pacific (Sasaki, 1985; Wolotira et al., 1993). The Alaskan population ranges from the Bering Sea to northern British Columbia; some mixing between the two stocks occurs near southern Vancouver Island (McDevitt, 1990; Kimura et al., 1998; Hanselman et al., 2011a). Alaskan sablefish are managed as a single population due to their widespread migration within this region (Heifetz and Fujioka, 1991; Kimura et al., 1998; Hanselman et al., 2011a). Sablefish spawn offshore at depths up to 500 m, with an approximate peak spawning date during February–March in Alaska (Mason et al., 1983; Wing and Kamikawa, 1995; Wing, 1997) and a peak abundance of eggs occurring during late February (Doyle and Mier, 2016). While there is little information on the incubation time and depth of Alaskan sablefish eggs, studies on other stocks indicate that their eggs incubate for several weeks, during which time they increase in density and sink to depths of 400–1000 m

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before hatching (Mason et al., 1983; Kendall and Matarese, 1987). Newly hatched larvae move to the surface and remain neustonic as they drift toward nearshore waters, which serve as important nursery habitat for Alaskan sablefish (Sasaki, 1985; Kendall and Matarese, 1987; Rutecki and Varosi, 1997b; Wing, 1997). Alaskan young-of-year sablefish peak in abundance during late May (Doyle and Mier, 2016) and are believed to reach nearshore settlement areas during summer and fall (Rutecki and Varosi, 1997a,b). Juveniles remain in these nearshore nursery sites until roughly age 2, at which point they migrate back offshore (Rutecki and Varosi, 1997b; Maloney and Sigler, 2008). They reach their adult habitat and recruit to the fishery at around age 4 (Maloney and Sigler, 2008).

1.2. Pacific ocean perch

POP inhabit demersal shelf and slope habitat throughout the North Pacific, but catches are most abundant in the GOA and Aleutian Islands (Carlson and Haight, 1976; Allen and Smith, 1988). POP in the GOA consist of genetically distinct subpopulations that occur at small geographic scales, indicating that larval dispersal is limited (Palof et al., 2011; Kamin et al., 2013). Female POP are ovoviparous, maintaining fertilized eggs internally until hatching and larval parturition (Gunderson, 1971). Parturition occurs during April and May in gullies, canyons, and along the slope at depths of 500–700 m (Paraketsov, 1963). POP larvae are pelagic, remaining primarily within the upper 100 m of the water column before settling in their demersal nearshore nursery habitat, which they likely reach within a year and may reach by their first autumn (Carlson and Haight, 1976; Doyle and Mier, 2016). Their preferred nearshore habitat consists of rough substrata and complex structure comprised of boulders, corals, and sponges (Carlson and Haight, 1976; Rooper et al., 2007). They

remain in their nursery habitat until roughly age 3, at which point they begin to migrate offshore toward their adult habitat (Carlson and Haight, 1976; Carlson and Straty, 1981). They reach their adult habitat in gullies and along the continental slope by age 6 (Carlson and Haight, 1976; Gunderson, 1977).

1.3. Study area

The GOA continental shelf varies in width from 5 km to 200 km; it is generally narrower in the eastern GOA and broader to the west (Weingartner, 2007). Depth on the shelf predominantly ranges over 150–200 m, and the shelf-break occurs at a depth of 200–300 m (Weingartner, 2007).

Ocean circulation in the GOA is primarily influenced by two main current systems: the Subarctic Gyre over the slope and basin and the Alaska Coastal Current (ACC) along the coast (Fig. 1; Stabeno et al., 2004). The Subarctic Gyre system is comprised of three currents that combine to create a cyclonic circulation pattern: the North Pacific Current, the Alaska Current, and the Alaskan Stream (Stabeno et al., 2004). Transport of water in this current system is mostly driven by cyclonic winds, which are associated with downwelling-favorable conditions along Alaska's southern coastline (Stabeno et al., 2004). During spring and summer months, these downwelling conditions weaken and are accompanied by intermittent, variable upwelling episodes (Stabeno et al., 2004).

The ACC flows counter-clockwise along the GOA coastline before flowing into the Bering Sea through the Aleutian passes, and is driven by a combination of alongshore winds and freshwater discharge into the GOA (Royer, 1982; Stabeno et al., 2004; Weingartner et al., 2005). The prominence of its freshwater core is reduced during spring and summer as water spreads offshore due to relaxed downwelling or upwelling (Stabeno et al., 2004).

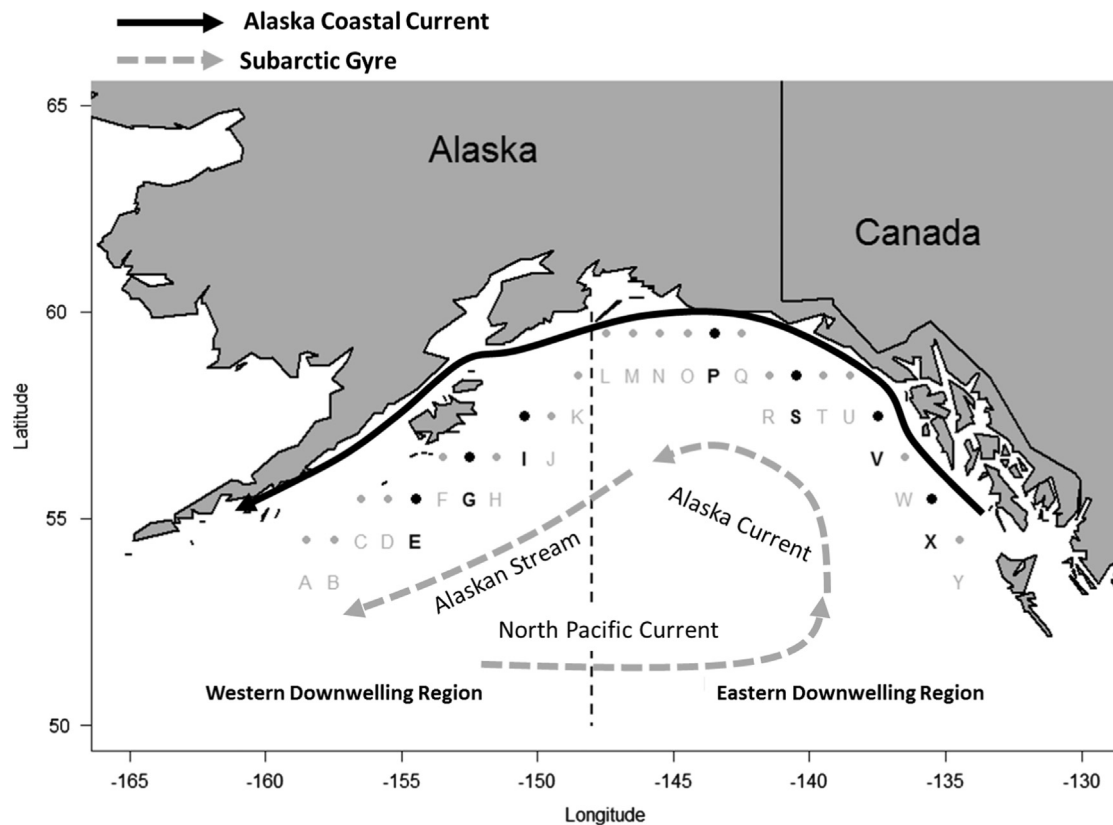


Fig. 1. Map showing GOA current systems (arrows), downwelling index locations used in analysis (points), and downwelling regions. The dotted line marks the division between eastern and western downwelling regions identified in this study (see Fig. 4). Black points (labeled in bold) denote locations used as subsamples to represent regional downwelling variability.

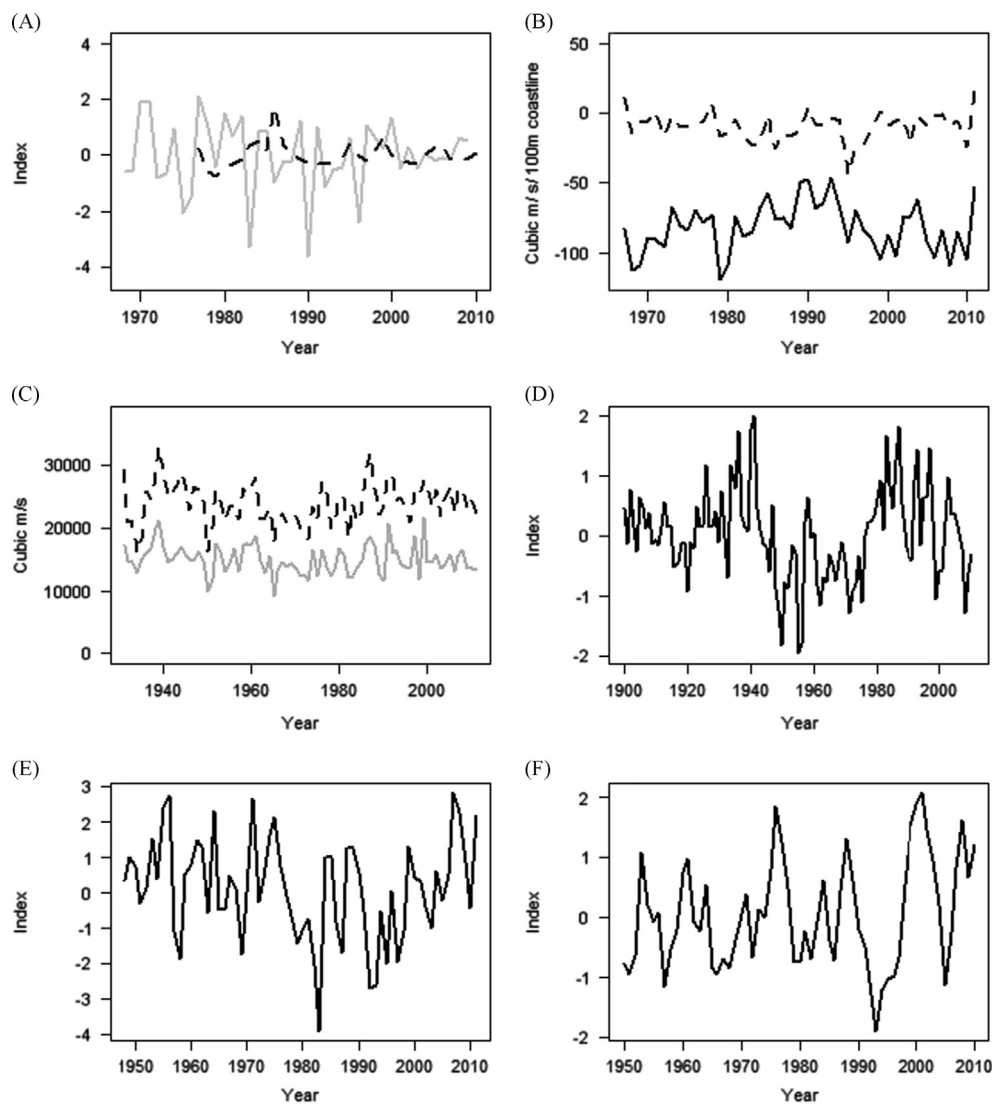


Fig. 2. Annual time series of variables used in analysis. Panels refer to recruitment (Ricker model residuals; a), upwelling (b), freshwater discharge (c), PDO (d), NOI (e), and NPGO (f). For recruitment, the solid line represents sablefish (years 1968–2009) and the dotted line represents POP (years 1977–2010). For upwelling-favorable winds and discharge, solid lines represent the eastern GOA while dotted lines represent the western GOA (GOA total for freshwater discharge).

Regional environmental conditions in the GOA may be related on interannual time scales to large-scale climate oscillations, including those indexed by the Northern Oscillation Index (NOI; Schwing et al., 2002), the North Pacific Gyre Oscillation (NPGO; Di Lorenzo et al., 2008), and the Pacific Decadal Oscillation (PDO; Mantua et al., 1997). The NOI is based on differences in sea level pressure anomalies (SLPa) between the North Pacific High (NPH) and a low-pressure region near Darwin, Australia (Schwing et al., 2002). In the GOA, positive NOI values reflect lower sea level pressure toward the east, higher sea level pressure toward the west, cooler SST, and stronger upwelling-favorable winds (Schwing et al., 2002). The NPGO is defined as the 2nd principal component of variability in sea surface height anomalies in the northeast Pacific Ocean (Di Lorenzo et al., 2008). Positive NPGO phases are associated with stronger upwelling within the Subarctic Gyre and stronger downwelling along the GOA coast (Di Lorenzo et al., 2008). The PDO is defined as the leading principal component of sea surface temperature variability in the North Pacific (Mantua et al., 1997). Positive PDO values reflect enhanced downwelling-favorable winds, warmer sea surface temperatures, and greater precipitation in the GOA (Cayan and Peterson, 1989; Trenberth, 1990; Salmon, 1992; Trenberth and Hurrell, 1994; Mantua et al., 1997).

1.4. Goals and rationale

Sablefish and POP recruitment in Alaska both have very high interannual variability (Fig. 2). No reliable stock–recruit relationships have been identified for either species, suggesting that environmental factors play a dominant role in determining their year-class strength (Hanselman et al., 2011a,b; Shotwell et al., 2014). Sablefish recruitment in Alaska has been linked to some large-scale climate and oceanographic features, including the Aleutian Low and the Polar Front (Shotwell et al., 2014). However, relationships between recruitment and regional-scale environmental indices in the GOA remain largely unexplored for both species, and may contribute insight into the mechanisms linking recruitment to large-scale oceanographic features (Shotwell et al., 2014).

One potentially important factor affecting variability in the recruitment of slope-spawning species such as sablefish and POP is their need to cross the shelf to reach nearshore nursery habitats after hatching. In general, potential cross-shelf advection mechanisms in the GOA may include onshore advection in deep waters during upwelling and relaxed downwelling, retention and flow modification by features such as eddies, bathymetric steering at depth, and shoreward advection at the surface during downwelling-favorable

conditions (Weingartner, 2007; Bailey et al., 2008). While mechanisms operating at depth are unlikely to apply to sablefish or POP larvae, it is likely that the GOA's cyclonic currents and wind conditions influence larval advection of both species across the shelf to their juvenile nursery habitats, as they do for other deep-spawning fish species in Alaska (e.g., Bailey et al., 2008; Doyle et al., 2009).

The main goal of this study is to investigate potential relationships between large-scale climate variables (indicated by the PDO, the NPGO, and the NOI), regional environmental variables (indicated by downwelling and freshwater discharge indices), and recruitment of sablefish and POP in the GOA. We focus on downwelling and freshwater discharge because cross-shelf transport of larvae at the surface is likely to be strongly affected by surface Ekman transport and alongshelf currents. By focusing on variables potentially related to advection, we hope to gain an understanding of some of the mechanistic implications behind potential environmental drivers of sablefish and POP recruitment. Specifically, we propose that their recruitment may be enhanced if shoreward advection is increased by strong downwelling conditions during the period of larval drift. If this is the case, recruitment of both species may show a positive relationship to downwelling strength during this time. A strong ACC, in contrast, may act as a barrier to shoreward advection when coupled with downwelling conditions that intensify the front between this freshwater-core current and surrounding seawater, and may therefore weaken recruitment if larvae are lost through alongshore advection toward unfavorable areas. Finally, given that the population structure of POP suggests limited lifetime dispersal, enhanced alongshore transport may reduce their larval survival and subsequent recruitment through enhanced advection toward unfavorable habitat.

Our climate indices were chosen primarily based on their relationships to oceanographic circulation in the GOA. While each of these three climate indices is associated with winds and circulation, weak to moderate correlations among the indices suggest that each captures unique aspects of the variability in circulation and transport in the GOA; therefore, we included all three indices in the analysis. The NOI oscillates at shorter interannual time scales than the other two indices, which operate on interdecadal time scales; furthermore, the NPGO and the PDO are statistically independent of each other and thus relate to different sources of variability in winds and circulation (Mantua et al., 1997; Schwing et al., 2002; Di Lorenzo et al., 2008; Chhak et al., 2009).

While ocean circulation was our primary motivation for choosing these climate variables, each one is linked to ecological measures of variability as well. The NPGO is correlated with primary productivity throughout the northeast Pacific, including the GOA (Di Lorenzo et al., 2008). The PDO has been linked to variability in productivity at several trophic levels in the GOA and elsewhere (e.g., Venrick et al., 1987; Brodeur et al., 1996; Anderson and Piatt, 1999). The NOI has also been linked to changes in biological time series, though most studies have focused on the North American west coast (e.g., Kahru and Mitchell, 2000; Mackas et al., 2001; Schwing et al., 2002). Thus, each of these indices may be useful for exploring other ecological forcing mechanism beyond advection.

We further propose that conditions in the eastern (E) GOA may have a more significant impact on recruitment than conditions in the western (W) GOA, especially for sablefish. Although POP and sablefish recruitment are estimated for the entire GOA and for the combined GOA and Bering Sea region, respectively, recruitment may nevertheless be affected by regional variability in downwelling and freshwater discharge. The broader continental shelf in the W GOA (Weingartner et al., 2002, 2009) may make shoreward advection of larvae more difficult in this region due to the greater distance required to travel from shelf-break to shore, whereas the narrower shelf in the E GOA exhibits greater cross-shelf exchange (Stabeno et al., 2016). This may cause downwelling in one region to be more important to overall recruitment than downwelling in another region.

While there is strong alongshore connectivity in the ACC (Royer, 1982), it is not a continuous feature, with a break occurring between Cross Sound and Yakutat (Stabeno et al., 2016). Additionally, most sablefish spawning likely occurs in the E GOA (Heifetz and Fujioka, 1991; Wing and Kamikawa, 1995; Sigler et al., 2001; Maloney and Sigler, 2008). Therefore, we examined downwelling-favorable winds and freshwater discharge separately for each region and used regional-scale indices in addition to large-scale climate indices in our analyses.

1.5. Hypotheses

To form *a priori* hypotheses about the relationship between environmental variables and recruitment processes, it is important to select appropriate spatial and temporal periods over which the hypothesized mechanisms operate (e.g., Mueter et al., 2005). Sablefish larvae are present on the shelf during April and May (Doyle and Mier, 2016), which coincides with the timing of larval parturition for POP (Gunderson, 1971, 1972, 1977). It is likely during this time period that downwelling-favorable winds at the shelf break play a role in enhancing onshore advection for both species. Since the ACC is located close to shore (Royer, 1982; Stabeno et al., 2004), freshwater discharge may play a role in the advection of sablefish larvae during late summer and fall (approximately July–September), just prior to their nearshore settlement (Rutecki and Varosi, 1997a,b). While the timing for POP settlement is unknown, their larvae may reach the ACC during the same approximate time period as sablefish larvae if sablefish hatching and POP parturition occur along the slope during the same approximate time period.

Based on these considerations, we specifically tested the following *a priori* hypotheses:

H_{A1}. Recruitment is negatively related to upwelling-favorable winds in the E GOA during mid-spring, which in turn are positively related to winter NOI and negatively related to winter PDO.

H_{A2}. Recruitment is negatively related to total freshwater discharge during summer or fall, which in turn is positively related to winter PDO.

H_{A3}. Recruitment shows a negative relationship to the NPGO.

H_{A1} and H_{A2} apply to each species, but H_{A3} was only hypothesized for POP due to our supposition that their population structure indicates sensitivity to alongshelf transport as indicated by the NPGO index. Because sablefish are believed to form a single coastwide population connected through active migration, we do not hypothesize any direct relationship between the NPGO and sablefish recruitment. An annual NPGO index was used in the analysis because, unlike our other indices, the NPGO index showed a coherent pattern across seasons. Each of these hypotheses is consistent with the overarching hypothesis that large-scale climate conditions affect the survival of young-of-year larvae, and their subsequent recruitment, through variability in cross-shelf advection toward suitable nursery grounds.

2. Materials and methods

We used a structural equation modeling (SEM; Bollen, 1989) approach to examine sequential relationships between recruitment, regional environmental variables, and large-scale climate variables at different seasonal lags. SEM is a confirmatory modeling technique that allows simultaneous testing of multiple hypothesized relationships, referred to as “paths” in SEM, between variables as separate steps within a complex model (Bollen, 1989; Ullman et al., 2007; Grace et al., 2010). Each path leads from an exogenous (explanatory) variable to an endogenous (response) variable; variables fulfilling both roles (i.e., those occupying the middle step

within the model) are often referred to as “intermediary” variables (Bollen, 1989; Ullman et al., 2007; Grace et al., 2010). Our SEM analysis was divided into two phases: one testing our *a priori* hypotheses, outlined above, followed by an exploratory phase testing relationships beyond those hypothesized beforehand.

2.1. Data

We compiled monthly or annual time series of biological (response) and physical (explanatory) variables spanning at least three decades (see Table A.1 in Appendix; Fig. 2). The abundances of recruits for each species were obtained from age-structured models used in each species' stock assessment (Hanselman et al., 2011a,b) and were used to examine factors that may affect early life survival and subsequent recruitment to the adult stock.

Although there is little evidence for a stock–recruitment relationship in these species and uncertainty about its functional form (Hanselman et al., 2011a,b), we considered it prudent to remove potential density-dependence prior to analysis. To adjust for potential effects of spawner abundance on recruitment, we fit a Ricker model for each species as follows:

$$\log(R_{t+2}/S_t) = \alpha + \beta S_t + \varepsilon_t,$$

where R_{t+2} denotes age 2 recruit abundance in year $t+2$, S_t denotes spawning biomass in year t , α and β denote the productivity and carrying capacity parameters of the Ricker stock–recruitment relationship, respectively, and ε_t denotes residual errors. The time series of residuals from each Ricker model were used as the response variable (hereafter referred to as “recruitment”) in all SEM analyses as a measure of anomalies in (log-transformed) survival from spawning to recruitment. Repeating the analyses with log-transformed raw recruitment time series did not affect our overall conclusions, indicating that results are robust to the nature of the stock–recruitment relationship. This was expected given that the log-transformed recruit abundance and Ricker model residual time series were highly correlated for both sablefish ($r=0.97$; $p<0.001$) and POP ($r=0.92$; $p<0.001$).

Monthly freshwater discharge estimates ($\text{m}^3 \text{s}^{-1}$) are based on a model incorporating precipitation, air temperature, and glacial melt (Royer, 1982; Royer and Grosch, 2006). These estimates are provided for two coastal drainage regions: discharge in the E GOA (east of Yakutat) and discharge in the central (C) GOA (between Yakutat and Seward); estimates are also provided for total discharge near Seward, which combines the E and C GOA indices and provides a measure of variability in the total volume flow of the ACC downstream of Seward (Royer, 1982; Weingartner et al., 2005). Although the three indices are moderately to highly correlated (E GOA vs C GOA: $r=0.66$; E GOA vs total: $r=0.74$; C GOA vs. total: $r=0.74$; all $p<0.001$), we used both E GOA and total discharge in the analysis to account for potential relationships specific to E GOA discharge.

The downwelling index ($\text{m}^3 \text{s}^{-1}/100 \text{ m}$ coastline) is described as the onshore/offshore component of Ekman transport, and is calculated based on geostrophic wind stress and the orientation of the adjacent coastline (PFEL, 2011). Index time series were obtained for 25 selected locations positioned along the shelf-break in the GOA at 1° longitudinal intervals (Fig. 1; Table 1). Positive values denote upwelling-favorable winds and negative values denote downwelling-favorable winds. Downwelling-favorable winds at the shelf-break may not necessarily reflect local downwelling at the coast; however, measuring vertical transport near the coast was not the purpose of this index in our study. Rather, we used this index as an indicator of broad-scale, cross-shelf Ekman transport toward shore, which may be conducive to shoreward advection of slope-spawned larvae toward their nursery areas. We examined downwelling separately for the E and W GOA because preliminary analyses indicated that each of these two regions exhibits a unique downwelling pattern with no coherent gulfwide pattern.

Table 1

Latitude, longitude, and coast angle (used to calculate downwelling index) of downwelling locations.

Station	Latitude	Longitude	Coast angle (deg.)
A	54.5	−158.5	60
B	54.5	−157.5	60
C	55.5	−156.5	60
D	55.5	−155.5	60
E	55.5	−154.5	60
F	56.5	−153.5	60
G	56.5	−152.5	60
H	56.5	−151.5	60
I	57.5	−150.5	60
J	57.5	−149.5	60
K	58.5	−148.5	60
L	59.5	−147.5	90
M	59.5	−146.5	90
N	59.5	−145.5	90
O	59.5	−144.5	90
P	59.5	−143.5	90
Q	59.5	−142.5	120
R	58.5	−141.5	120
S	58.5	−140.5	120
T	58.5	−139.5	120
U	58.5	−138.5	120
V	57.5	−137.5	140
W	56.5	−136.5	140
X	55.5	−135.5	140
Y	54.5	−134.5	140

2.2. Analytical approach

Before SEM analysis, we characterized spatial and/or seasonal variability in each environmental variable to select appropriate spatial and seasonal averages for the analysis. For each environmental variable, the full time series (see Table A.1 in Appendix) was used to elucidate seasonal and spatial patterns. For use in the SEM analysis, time series were truncated to only include years that had data for all variables including recruitment (1977–2010 for POP; 1968–2009 for sablefish). Freshwater discharge was provided for two relatively large regions and could not be analyzed at a smaller spatial scale. Climate indices (PDO, NOI, and NPGO) represent large-scale phenomena; hence, only temporal variability was analyzed for these variables.

Because of large variability in downwelling-favorable winds along the GOA shelf break, we examined spatial patterns in upwelling using a hierarchical cluster analysis. Daily time series were compiled for each of the locations identified in Fig. 1. Temporal patterns of variability in these time series were compared among locations using pairwise Euclidean distances and Ward's minimum variance clustering (Ward, 1963). Results from this cluster analysis were mapped into dendrograms, wherein locations whose time series showed similar patterns of variability were grouped together. These were used to visually identify groups of spatially continuous locations with similar patterns in variability in downwelling-favorable winds, each of which was considered a downwelling region. Within each identified region, upwelling indices at representative locations close to the shelf break were averaged by region for further analyses.

A similar hierarchical cluster analysis was performed on the monthly time series of all environmental indices in order to identify times of year showing coherent patterns of variability, thus defining seasonal patterns. First, each index was averaged by month to form 12 interannual time series, one for each month. Temporal patterns of variability were then compared among months using pairwise Euclidean distances and Ward's minimum variance clustering (Ward, 1963). The resulting dendrograms were used to visually identify groups of months with similar interannual patterns of variability. Any consecutive months grouped together were considered a distinct

“season” for the purposes of further analysis. Conversely, individual months showing no or limited similarity to the preceding or following months were considered their own distinct season. This analysis was performed separately for each region of the downwelling index and for E and total GOA discharge indices. In the event that a break was found between “December” and “January” for a given variable, cluster analysis was repeated using a non-calendar year (e.g., November of year t through October of year $t+1$) to investigate whether or not this was merely an artifact from using a calendar year. If so, the non-calendar year was used.

To examine relationships between recruitment and potential drivers, we applied the SEM approach described in Bollen (1989) and Ullman et al. (2007), which can be briefly summarized as follows. First, a model outlining paths among climate variables, regional variables, and recruitment was specified. Next, each hypothesized path was translated into an equation from which parameters were estimated. These parameter estimates were used to construct a covariance matrix corresponding to the model, which was compared to the sample covariance matrix using a chi-square (χ^2) test. An insignificant χ^2 value reflects similar covariance matrices, indicating that the specified model is consistent with the observed data. Models with significant χ^2 values ($p < 0.05$) were therefore considered implausible and were ruled out as candidate models, while those with insignificant χ^2 values were considered plausible and examined further. Within a given model, each included path was tested for statistical significance separately.

While SEM allows for inclusion of latent variables, or variables for which no observed measurements are available (Bollen, 2002), such variables are seldom used in ecological analyses (Grace et al., 2010) and were not used here. Instead we used a variation of structural models in which latent variables are excluded:

$$\mathbf{y} = \mathbf{I}\mathbf{x} + \mathbf{B}\mathbf{y} + \boldsymbol{\zeta},$$

Where, if q is the number of endogenous variables and p is the number of exogenous variables, $\mathbf{y}$ denotes a $q \times 1$ vector of endogenous and intermediary variables (e.g., recruitment and upwelling); $\mathbf{x}$ denotes a $p \times 1$ vector of exogenous variables (e.g., PDO); $\mathbf{I}$ denotes a $q \times p$ matrix of regression coefficients between exogenous and endogenous variables; $\mathbf{B}$ denotes a $q \times q$ matrix of regression coefficients between endogenous and intermediary variables; and $\boldsymbol{\zeta}$ denotes a vector of error terms (Bollen, 1989). The placement of $\mathbf{y}$ on both sides of the equation is due to the ability of endogenous variables to predict one another in SEM when they serve as intermediary variables (Bollen, 1989). Within each matrix of regression coefficients, the coefficients corresponding to a given pair of variables can be set to 0 if no relationship between those variables is hypothesized, set to a previously hypothesized fixed value, or estimated (Bollen, 1989; Ullman et al., 2007). Thus, relationships between endogenous variables and intermediary variables can be established in the $\mathbf{B}$ matrix by estimating the regression coefficient between them rather than fixing it at 0 (Bollen, 1989; Ullman et al., 2007). To visually represent the proposed relationships between variables, SEM models were mapped into path diagrams (e.g., Fig. 3).

We used two separate approaches of SEM analysis for each species. The first approach was a hypothesis-testing phase, in which we tested our *a priori* hypotheses for each species. For this phase, we applied a two-year lag between pre-selected seasonal indices of explanatory variables and age 2 recruitment in order to model variability in year-class strength against environmental conditions effective during the hypothesized time periods at age 0.

To test our hypotheses, a series of alternative SEM models were compared for each species, including:

- 1) A full model containing all variables at appropriate lags that tested all applicable hypotheses simultaneously (Fig. 3),

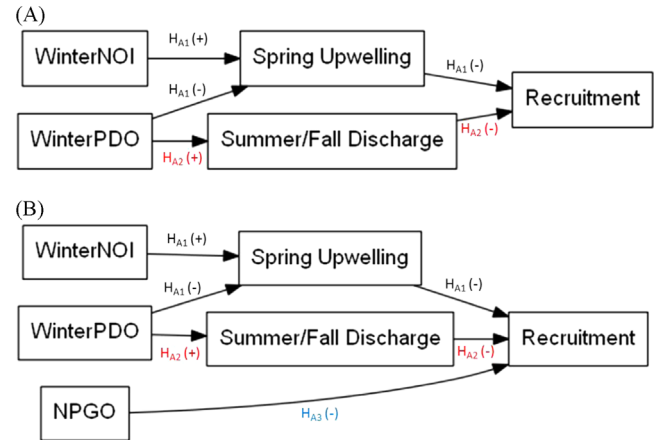


Fig. 3. Path diagrams showing hypothesized relationships between variables. Relationships are shown for sablefish (a) and POP (b). Arrows point from exogenous variables to endogenous variables. Edge labels denote the overarching hypothesis that a given path is part of as well as the hypothesized sign of that path.

- 2) Multiple reduced models testing all possible combinations of hypotheses at a time (e.g., H_{A1} and H_{A2} ; H_{A1} only, H_{A2} only, etc.), and
- 3) Variants of each model wherein the hypothesized seasonal time period for each variable was represented by alternate seasonal averages, if applicable. For example, spring downwelling was represented by April downwelling in one model and May downwelling in another.

Because H_{A1} and H_{A2} each propose multiple sequential relationships, these hypotheses were rejected unless all such relationships were found to be significant (at $\alpha=0.05$) within one or more plausible candidate structural models. For instance, if a plausible model contained a significant path between summer discharge and recruitment and an insignificant path between winter PDO and summer discharge, this model would be insufficient for rejecting the null hypothesis in favor of H_{A2} . Therefore, for each of the alternative hypotheses, the null hypothesis (H_0) is that none of the proposed relationships between variables are significant. To align the seasons identified in our cluster analyses with our hypotheses, we used downwelling during April and May and discharge during summer (July–September) and fall (October–November) to represent the hypothesized time periods during which these variables are related to recruitment of each species. For the PDO and the NOI, we used winter indices because large-scale wintertime conditions tend to relate to groundfish recruitment more strongly (Hollowed and Wooster, 1992), likely through delayed effects on spring and summer regional variables.

Because we found little support for these *a priori* hypotheses, we extended the analyses to include an exploratory phase. In the exploratory phase, recruitment was modeled against an extended range of seasonal lags for each variable throughout each species' early life history (age 0–2), testing different combinations of climate and regional explanatory variables. Different annual lags were applied in order to model recruitment against conditions during each year of their early life history; i.e., a two-year lag was applied to test conditions during age 0, a one-year lag was applied to test conditions during age 1, etc.

Although the hypothesis-testing and exploratory phases both tested relationships during age 0, our *a priori* hypotheses involved specific combinations of multiple complex relationships. In the exploratory phase, models were able to contain alternative combinations of relationships that were not tested during the hypothesis-testing phase. Although exploratory, these models still contained

assumptions of sequential causation between climate and regional variables; therefore, climate variables at each seasonal lag were always modeled with regional variables at concurrent or later seasonal lags. The large number of models tested in the exploratory phase contains an increased risk of Type I error. We address this by employing multiple criteria by which models were eliminated from the analysis. Implausible exploratory models (i.e., those with significant χ^2 values) were excluded, and remaining candidate models were compared using the small-sample Akaike Information Criterion (AICc); if the difference in AICc values (ΔAICc) was larger than 2, the model with the smallest AICc was chosen as the final model (Burnham and Anderson, 2002). If multiple alternative models had similar support ($\Delta\text{AICc} < 2$), the most parsimonious model was chosen. In the event that models with similar support also had the same number of parameters, all such models were equally considered. Consequently, all models with ΔAICc values > 2 relative to the smallest AICc value were eliminated from the analysis alongside models deemed implausible according to their χ^2 values, reducing the potential for spurious relationships. Additionally, we took into account incidents in which multiple final candidate models contained the same significant relationship, strengthening the evidence for an effect. The exploratory phase was performed in such a way that relationships during each year of each species' early life history were examined separately; i.e., models testing relationships during age 0 were only compared with other models testing relationships during age 0, and likewise for ages 1 and 2.

The statistical software package R, version 2.14.0 (R Development Core Team, 2011), was used for all analyses. Structural models were fit using the 'sem' package (Fox, 2006).

3. Results

3.1. Cluster analysis

Two distinct downwelling regions within the GOA were identified via cluster analysis: one for the E GOA and one for the W GOA, with a break point between locations K (147.5° west) and L (148.5° west; Fig. 1). The downwelling index in the E GOA indicated a stronger seasonal signal, with stronger downwelling-favorable conditions during fall and winter (Fig. 4). Mean winter values throughout the time series averaged $-168.27 \text{ m}^3 \text{ s}^{-1}/100 \text{ m}$ coastline for the E GOA, lower than the time series averages for any other seasons for this region. The season with the highest time series average in the E GOA was June, averaging $-1.44 \text{ m}^3 \text{ s}^{-1}/100 \text{ m}$ coastline throughout the time series. In contrast, the W GOA exhibited a weaker seasonal pattern, weaker wintertime downwelling-favorable winds, and a longer period of upwelling-favorable winds relative to the E GOA (Fig. 4). Mean seasonal index values for the W GOA were lowest during the winter/spring season (January–April), which averaged $-24.94 \text{ m}^3 \text{ s}^{-1}/100 \text{ m}$ coastline throughout the time series. The highest seasonal mean in the W GOA occurred during August, which averaged $13.55 \text{ m}^3 \text{ s}^{-1}/100 \text{ m}$ coastline throughout the time series. Time series averages in the W GOA were positive (upwelling-favorable) for all seasons between July and October.

For the locations representing the E GOA, nine distinct monthly groupings ("seasons") with respect to downwelling were identified (Fig. 5) because several months did not cluster with any other group. Because the "winter" season included November of year t through January of year $t+1$, the time series' first year (1967) had to be excluded from the sablefish analysis and only years 1968–2009 were used. For the locations representing the W GOA, eight distinct seasons were identified (Fig. 6). For E GOA discharge, four distinct seasons (not shown) were selected: early winter (November–January), late winter (February–March), April, and summer/fall (May–October; see Fig. A.1 in Appendix for time series). For total discharge, five seasons (not

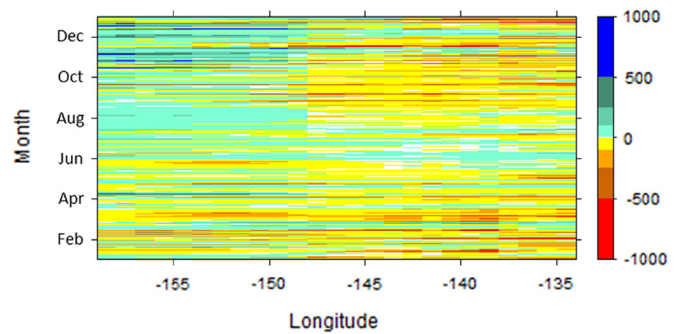


Fig. 4. Hovmöller diagram showing downwelling index values by month and longitude. Daily values are averaged across years 1967–2011 and longitude is on a 1-degree resolution from west to east. Warm colors (negative values) represent downwelling-favorable winds and cool colors (positive values) represent upwelling-favorable winds.

shown) were selected: winter (December–January), early spring (February–April), late spring (May–June), summer (July–September), and fall (October–November; see Fig. A.1 in Appendix for time series).

Seasons selected for the NOI included winter (September–April) and summer (May–August). For the PDO, three seasons were selected: winter (January–March), spring/summer (April–September), and late fall (October–December). The NPGO showed no obvious seasonal patterns; therefore, an annual mean NPGO index was used in all further analyses.

3.2. SEM analysis

3.2.1. Sablefish

In the hypothesis-testing phase for sablefish, both *a priori* hypotheses (H_{A1} and H_{A2}) were rejected. A total of 12 models were tested in the hypothesis-testing SEM phase; of these, 11 models had the insignificant χ^2 values required to be considered as candidate models. Candidate models testing both H_{A1} and H_{A2} simultaneously had less support than smaller models testing one of these hypotheses at a time (Table 2). The two best candidate models both tested H_{A2} only, but contained no paths that were individually significant. Among the remaining models, the most consistently significant paths were a negative correlation between winter PDO and winter NOI and a negative relationship between winter NOI and March (E GOA) upwelling-favorable winds. No models contained any significant paths between environmental variables and recruitment and the R^2 values for recruitment were generally smaller than 5%.

The exploratory analysis indicated that sablefish recruitment was negatively related to above average late winter and early spring discharge during both age 1 and age 2, and positively related to stronger July upwelling-favorable winds during age 1 (Fig. 7). Recruitment was unrelated to downwelling or discharge conditions during the larval and early juvenile phase (age 0). A total of 35 models were tested examining relationships during age 0; of these, 26 models had insignificant χ^2 values and the 10 best models had similar AICc values and the same number of parameters (see Table A.2 in Appendix). Only three significant paths were found within these models: a positive relationship between winter PDO and early spring (total) discharge, a negative relationship between winter PDO and summer/fall (E GOA) discharge, and a negative correlation between winter PDO and winter NOI. R^2 values for recruitment were all below 5% in these models.

Recruitment was positively related to July upwelling-favorable winds and negatively related to late winter discharge in the eastern GOA during age 1. Of the 42 models testing relationships during age 1, 34 models had insignificant χ^2 values; the 15 best models had similar AICc values and the same number of parameters (see Table A.3 in Appendix). Of these, 4 models showed a significant positive relationship

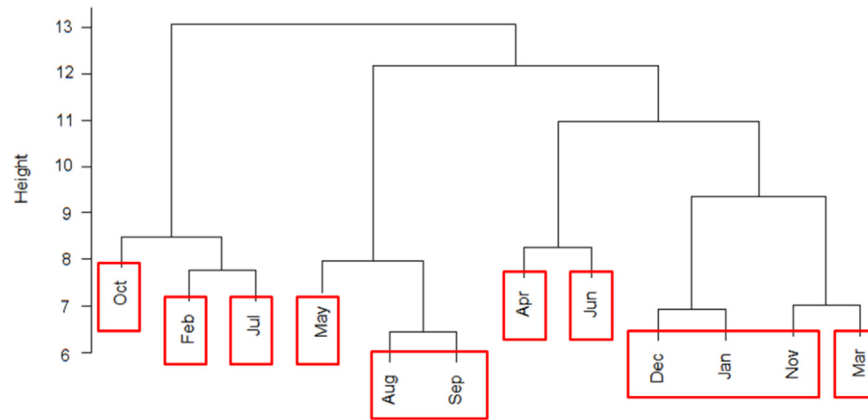


Fig. 5. Ward's minimum variance dendrogram resulting from cluster analysis of the monthly eastern GOA downwelling index. Boxes mark groups selected as distinct seasonal lags.

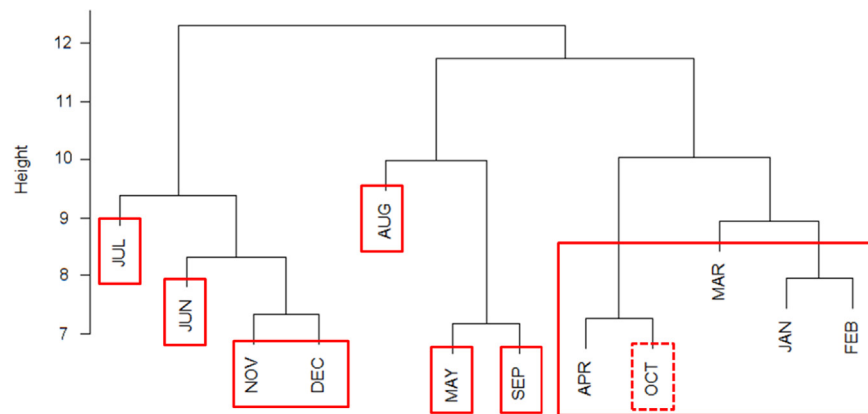


Fig. 6. Ward's minimum variance dendrogram resulting from cluster analysis of the monthly western GOA downwelling index. Boxes mark groups selected as distinct seasonal lags. The dotted box around October indicates its exclusion from the larger cluster containing it, due to its being nonconsecutive with those months.

between July (E GOA) upwelling-favorable winds and sablefish recruitment. One model showed a sequential set of significant relationships wherein winter PDO was positively related to late winter (E GOA) discharge, which was negatively related to recruitment. Six models showed no significant relationships.

Sablefish recruitment showed a negative relationship to total discharge during early spring of age 2. A total of 41 models were tested to examine relationships during age 2, 35 of which had insignificant χ^2 values. The 11 best models had similar AICc values and the same number of parameters (see Table A.4 in Appendix). One model showed a sequential set of significant relationships wherein winter PDO was positively related to early spring (total) discharge, which was negatively related to recruitment. Other models showed significant relationships between winter PDO and E GOA discharge during late winter and summer/fall, and 6 models showed no significant relationships.

3.2.2. POP

Though we found little support for our *a priori* hypotheses for POP during the hypothesis-testing SEM phase, there was support for some components of H_{A1} ; specifically, we found that recruitment of POP was positively related to stronger downwelling-favorable winds during late spring. Of the 35 models tested during this phase, 13 models had insignificant χ^2 values and were thus considered as candidate models. Three models contained a significant negative relationship between June (E GOA) upwelling-favorable winds and recruitment, and 2 models contained a significant negative relationship between June (W GOA) upwelling-favorable winds and

recruitment. A significant negative correlation between winter PDO and winter NOI was shown in 11 plausible models (Table 3). The best candidate model tested H_{A3} only and contained no significant paths.

The exploratory SEM phase indicated several significant relationships between POP recruitment and environmental variables at multiple lags (Fig. 8). Recruitment of POP was negatively related to stronger June upwelling-favorable winds in the western GOA during age 0. A total of 47 models were tested examining relationships during age 0, 33 of which had insignificant χ^2 values. Of these, the 9 best models had similar AICc values and the same number of parameters (see Table A.5 in Appendix). Two of these models showed a significant negative relationship between June (W GOA) upwelling-favorable winds and recruitment. March (E GOA) upwelling-favorable winds showed a negative relationship to the NPGO in one model and to winter NOI in another. In 5 models, no significant relationships were found.

POP were also found to be negatively associated with stronger June upwelling-favorable winds in both the eastern and western GOA during age 1, as well as with total discharge during late spring. A total of 49 models were tested examining relationships during age 1, 40 of which had insignificant χ^2 values. The 9 best models had similar AICc values and the same number of parameters (see Table A.6 in Appendix). Three of these models showed a significant negative relationship between June (E GOA) upwelling-favorable winds and recruitment, and 3 others showed a significant negative relationship between June (W GOA) upwelling-favorable winds and recruitment. March (E GOA) upwelling-favorable winds showed negative relationships to both the NPGO and winter NOI, and late spring (total)

Table 2

Summary statistics for SEM models from the sablefish hypothesis-testing phase. Summary statistics for individual paths are only included for plausible models. Arrows point from exogenous to endogenous variables; double-headed arrows denote correlations with no causal relationship hypothesized. ΔAICc is given relative to the model with the lowest AICc; K represents the number of parameters. Shaded cells indicate p -values below 0.05.

Hypotheses tested	Adjusted goodness-of-fit	Pr ($> \chi^2$)	ΔAICc	K	Path	Path p -value	Path β -estimate	Endogenous variable R^2	H_0 outcome
H_{A1}, H_{A2}	0.76	0.10	14.80	11	April (E GOA) Upwelling $\rightarrow$ Recruitment	0.31	−0.16	0.04	Failure to reject
					Summer (Total) Discharge $\rightarrow$ Recruitment	0.38	−0.14		
					Winter NOI $\rightarrow$ April (E GOA) upwelling	0.86	0.03	0.02	
					Winter PDO $\rightarrow$ April (E GOA) upwelling	0.56	−0.11		
					Winter PDO $\rightarrow$ Summer (Total) discharge	0.06	−0.28	0.08	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}, H_{A2}	0.52	0.00	29.20	10					Failure to reject
H_{A1}, H_{A2}	0.83	0.27	12.27	11	March (E GOA) Upwelling $\rightarrow$ Recruitment	0.55	−0.09	0.02	Failure to reject
					Summer (Total) Discharge $\rightarrow$ Recruitment	0.40	−0.13	0.15	
					Winter NOI $\rightarrow$ March (E GOA) upwelling	0.02	−0.43		
					Winter PDO $\rightarrow$ March (E GOA) upwelling	0.68	−0.07	0.08	
					Winter PDO $\rightarrow$ Summer (Total) discharge	0.06	−0.28		
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}, H_{A2}	0.80	0.20	13.13	11	May (E GOA) Upwelling $\rightarrow$ Recruitment	0.74	0.05	0.01	Failure to reject
					Summer (Total) Discharge $\rightarrow$ Recruitment	0.52	−0.10		
					Winter NOI $\rightarrow$ May (E GOA) upwelling	0.06	−0.34	0.08	
					Winter PDO $\rightarrow$ May (E GOA) upwelling	0.32	−0.18		
					Winter PDO $\rightarrow$ Summer (Total) discharge	0.06	−0.28	0.08	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}, H_{A2}	0.90	0.57	10.04	11	March (E GOA) Upwelling $\rightarrow$ Recruitment	0.71	−0.06	0.00	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.87	0.02		
					Winter NOI $\rightarrow$ March (E GOA) upwelling	0.02	−0.43	0.15	
					Winter PDO $\rightarrow$ March (E GOA) upwelling	0.68	−0.07		
					Winter PDO $\rightarrow$ Fall (Total) discharge	0.18	−0.21	0.04	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}, H_{A2}	0.85	0.33	11.71	11	April (E GOA) Upwelling $\rightarrow$ Recruitment	0.39	−0.13	0.02	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.93	0.01		
					Winter NOI $\rightarrow$ April (E GOA) upwelling	0.86	0.03	0.02	
					Winter PDO $\rightarrow$ April (E GOA) upwelling	0.56	−0.11		
					Winter PDO $\rightarrow$ Fall (Total) discharge	0.18	−0.21	0.04	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}, H_{A2}	0.86	0.41	11.10	11	May (E GOA) Upwelling $\rightarrow$ Recruitment	0.61	0.08	0.01	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.74	0.05		
					Winter NOI $\rightarrow$ May (E GOA) upwelling	0.06	−0.34	0.08	
					Winter PDO $\rightarrow$ May (E GOA) Upwelling	0.32	−0.18		
					Winter PDO $\rightarrow$ Fall (E GOA) upwelling	0.18	−0.21	0.04	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}	0.92	0.50	4.07	8	March (E GOA) Upwelling $\rightarrow$ Recruitment	0.69	−0.06	0.00	Failure to reject
					Winter NOI $\rightarrow$ March (E GOA) upwelling	0.02	−0.43	0.15	
					Winter PDO $\rightarrow$ March (E GOA) upwelling	0.68	−0.07		
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}	0.90	0.43	4.37	8	April (E GOA) Upwelling $\rightarrow$ Recruitment	0.39	−0.13	0.02	Failure to Reject
					Winter NOI $\rightarrow$ April (E GOA) upwelling	0.86	0.03	0.02	
					Winter PDO $\rightarrow$ April (E GOA) upwelling	0.56	−0.11		
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	
H_{A1}	0.88	0.36	4.74	8	May (E GOA) Upwelling $\rightarrow$ Recruitment	0.65	0.07	0.00	Failure to Reject
					Winter NOI $\rightarrow$ May (E GOA) upwelling	0.06	−0.34	0.08	
					Winter PDO $\rightarrow$ May (E GOA) upwelling	0.32	−0.18		
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.00	−0.57	NA	

H_{A2}	1.00	0.91	0.00	5	Fall (Total) Discharge → Recruitment Winter PDO → Fall (Total) discharge	0.82 0.18	0.04 -0.21	0.00 0.04	Failure to Reject
H_{A2}	0.99	0.71	0.12	5	Summer (Total) Discharge → Recruitment Winter PDO → Summer (Total) discharge	0.48 0.06	-0.11 -0.28	0.01 0.08	Failure to Reject

discharge showed a significant positive relationship to recruitment in one model.

POP recruitment was unrelated to any environmental variables during age 2. Out of 47 models tested, 26 had insignificant χ^2 values and the 2 best models had similar AICc values and the same number of parameters (see Table A.7 in Appendix). Neither of these models showed any significant relationships between variables tested, and all R^2 values for recruitment were below 5%.

4. Discussion

4.1. A priori hypotheses

The rejection of all *a priori* hypotheses suggests that our chosen climate variables do not relate to recruitment through intermediary effects on our chosen regional variables at the hypothesized seasonal lags. Sablefish recruitment was unrelated to the environmental factors that we used as indicators for variability in cross-shelf and along-shelf transport during age 0. This may suggest that variability in larval advection toward nursery sites does not have a significant impact on their recruitment or that larval advection is not adequately represented by the indices chosen. POP recruitment was positively related to downwelling-favorable winds during June, but none of the hypotheses tested were accepted in their entirety.

4.2. Exploratory analyses

4.2.1. Sablefish

While we found little support for our *a priori* hypotheses, the exploratory analysis revealed some statistically significant relationships for both species that may reflect environmental influences on recruitment during juvenile stages, including a positive relationship between sablefish recruitment and July upwelling-favorable winds in the E GOA during age 1. The mechanism behind this relationship is unlikely to be related to advection, as sablefish reach their nursery sites well before this time period (Rutecki and Varosi, 1997a,b). Reduced downwelling in the summer may result in increased fall production along the E GOA coast (Waite and Mueter, 2013), which could benefit young sablefish in their coastal nursery grounds.

An analysis of the relationship between our sablefish recruitment index and seasonal chlorophyll-*a* indices for 1998–2009 from Waite and Mueter (2013) showed a positive relationship between recruitment and chlorophyll-*a* levels in the western GOA shelf region during fall at age 1 ($r=0.60$, $p=0.04$). However, none of the chlorophyll-*a* indices from Waite and Mueter (2013), including their eastern GOA index, were significantly correlated with our index of July (E GOA) upwelling-favorable winds. This suggests that the relationship between sablefish recruitment and July upwelling-favorable winds found in this study is unrelated to the relationship between summer upwelling and fall chlorophyll-*a* abundance found by Waite and Mueter (2013). The relationship between sablefish recruitment and July (E GOA) upwelling-favorable winds may be spurious or may result from unknown mechanisms. However, the correlation between July (E GOA) upwelling-favorable winds and W GOA onshelf chlorophyll-*a*, while insignificant, does show the expected positive sign ($r=0.01$; $p=0.98$). Its insignificance may simply be due to the limited length of the time series available for chlorophyll-*a*, resulting in low power for statistical tests of significance. In addition, the on-shelf division between the E and W GOA chlorophyll-*a* regions shown by Waite and Mueter (2013) occurs near the division between our two downwelling regions ($\sim 148^\circ\text{W}$); therefore, it is unlikely that the lack of statistical significance between our upwelling index and their chlorophyll-*a* index is due to mismatched regional divisions. Furthermore, while it is possible that the relationship between July (E GOA) upwelling-favorable winds and sablefish recruitment is spurious, its

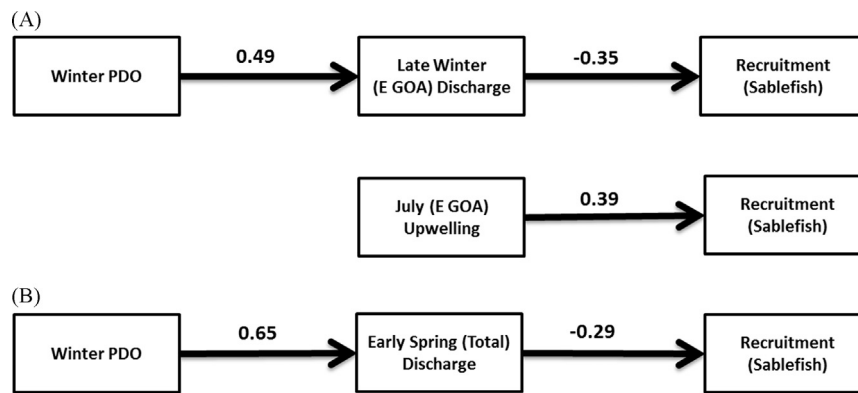


Fig. 7. Path diagrams mapping significant paths to sablefish recruitment found in candidate models from exploratory phase of SEM analysis. Only significant paths between environmental variables and recruitment are represented, including significant sequential paths. Diagrams are shown for models testing relationships during age 1 (a) and age 2 (b); no significant paths to recruitment were found during age 0. Arrows point from exogenous to endogenous variables. Numbers denote parameter estimates.

prevalence among the candidate models testing relationships between recruitment and environmental conditions during age 1 suggests that a more in-depth examination of the relationship between July upwelling-favorable winds, summer and fall productivity, and sablefish recruitment is warranted, perhaps when a longer chlorophyll time series is available.

The sequential relationships between winter PDO, late winter/early spring freshwater discharge during ages 1 and 2, and sablefish recruitment may also be due to productivity-related mechanisms. The intensified Aleutian Low associated with a positive PDO phase enhances onshore flow of moist air in the North Pacific (Cayan and Peterson, 1989; Trenberth, 1990; Trenberth and Hurrell, 1994; Mantua et al., 1997). This can lead to enhanced precipitation, earlier and enhanced water column stability, and earlier phytoplankton blooms during spring (Cayan and Peterson, 1989; Trenberth, 1990; Gargett, 1997; Weingartner et al., 2005; Henson, 2007). These early blooms tend to be more intense and shorter-lived, as they deplete nutrients more quickly than less intense blooms; furthermore, stronger water column stability throughout spring and summer can reduce primary and secondary productivity due to reduced mixing with nutrient-rich waters (Freeland et al., 1997; Eslinger et al., 2001; Henson, 2007). Because zooplankton reproduce more slowly than phytoplankton, this can result in underutilization of the intense, brief productivity associated with earlier blooms and reduced zooplankton biomass (Eslinger et al., 2001). Although the relationship between water column stability and primary productivity is more complex for the GOA shelf than for inner bays like Prince William Sound (PWS), both regions exhibit lower primary productivity during years with earlier, stronger, shallower stratification (Eslinger et al., 2001; Weingartner et al., 2005; Henson, 2007; Miller et al., 2012). Similarly, enhanced mixing near Cross Sound results in greater production on the nearby shelf (Staben et al., 2016). It is plausible that these results indicate that a positive PDO phase, which corresponds to warmer temperatures and enhanced precipitation, leads to greater freshwater discharge during late winter and early spring, and that this results in earlier water column stability, shorter, underutilized phytoplankton blooms, and reduced sablefish recruitment. Furthermore, the observation that freshwater discharge is related to recruitment at a slightly later lag during age 2 compared to age 1 is consistent with the observation that spring blooms typically occur later on the shelf than in enclosed coastal bays (Eslinger et al., 2001; Weingartner et al., 2005), and that age 2 is around when sablefish begin to leave coastal bays for shelf waters (Rutecki and Varosi, 1997b).

The indirect relationship between winter PDO and sablefish recruitment may also reflect mechanisms unrelated to freshwater discharge that our analysis did not explore. Population dynamics are often better explained by large-scale climate drivers than by local environmental

variables due to the tendency of wild populations to be affected by multiple factors that are not captured by a single local index (Hallett et al., 2004). This was not the case in our analysis, where both climate and regional indices were significantly related to recruitment. However, it is still possible that PDO is related to sablefish recruitment through complex mechanisms involving several environmental factors that include, but are not limited to, freshwater discharge.

4.2.2. POP

The observed negative relationship between POP recruitment and June upwelling-favorable winds in both the E GOA and W GOA during age 0 provides some support for our supposition that onshelf Ekman transport is conducive to POP recruitment, although it occurred later than our hypothesized larval advection period. Our hypothesis was based on the timing of parturition, which occurs during April and May for POP. However, the precise duration of the juvenile pelagic stage is unknown for POP, and it is plausible that it lasts beyond June (Carlson and Haight, 1976; Doyle and Mier, 2016).

POP recruitment was also negatively related to June upwelling-favorable winds during age 1. While this may be indicative of a mechanism acting during both age 0 and age 1 but unrelated to larval advection, it may also reflect different mechanisms at different ages. Throughout much of the year, strong downwelling winds transport zooplankton from oceanic origins toward shore (Cooney, 1986, 1988). Shotwell et al. (2014), after linking sablefish recruitment to advection of oceanic properties associated with the Polar Front, proposed a mechanism in which cool air from the Arctic enhances both primary and secondary production in central North Pacific waters, which are then advected onto the shelf during spring. Zooplankton and juvenile fish in PWS have exhibited offshore isotopic carbon signatures that vary interannually, indicating that planktivorous organisms in nearshore surface waters rely somewhat on a varying supply of productive waters from offshore that are likely transported to the shelf through onshore transport processes such as downwelling (Eslinger et al., 2001). Years with high cross-shelf mixing are also associated with higher zooplankton abundances in the northern GOA (de Sousa et al., 2016). It is possible that stronger downwelling-favorable winds at the slope enhance advection of age 0 POP toward shore while simultaneously transporting waters abundant in phytoplankton and/or zooplankton, enhancing prey availability to age 1 POP already located close to shore.

The observed positive relationship between POP recruitment and late spring (total) discharge may also be related to enhanced prey availability on the shelf. It is possible that higher discharge during late spring corresponds to longer winters, later stratification, and later spring blooms. In contrast with the dynamics of early spring blooms discussed earlier, years in which stratification occurs later in

Table 3

Summary statistics for SEM models from the POP hypothesis-testing phase. Summary statistics for individual paths are only included for plausible models. Arrows point from exogenous to endogenous variables; double-headed arrows denote correlations with no causal relationship hypothesized. $\Delta AICc$ is given relative to the model with the lowest AICc; K represents the number of parameters. Shaded cells indicate p -values below 0.05.

Hypotheses tested	Adjusted goodness-of-fit	Pr ($> \chi^2_1$)	$\Delta AICc$	K	Path	Path p -value	Path β -estimate	Endogenous variable R^2	H_0 outcome
All	0.58	0.00	39.47	13					Failure to reject
All	0.56	0.00	40.92	13					Failure to reject
All	0.60	0.00	40.11	13					Failure to reject
All	0.59	0.01	38.62	13					Failure to reject
All	0.58	0.00	41.09	13					Failure to reject
All	0.71	0.10	30.72	13	April (E GOA) Upwelling $\rightarrow$ Recruitment	0.52	-0.11	0.02	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.65	-0.08		
					NPGO $\rightarrow$ Recruitment	0.83	0.04		
					Winter NOI $\rightarrow$ April (E GOA) upwelling	0.76	0.06	0.01	
					Winter PDO $\rightarrow$ April (E GOA) upwelling	0.87	-0.03		
					Winter PDO $\rightarrow$ Fall (Total) Discharge	0.07	-0.30	0.09	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.01	-0.54	NA	
All	0.67	0.05	32.96	13					Failure to reject
All	0.74	0.17	29.10	13	June (E GOA) Upwelling $\rightarrow$ Recruitment	0.04	-0.33	0.12	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.56	-0.09		
					NPGO $\rightarrow$ Recruitment	0.86	-0.03		
					Winter NOI $\rightarrow$ June (E GOA) upwelling	0.14	-0.29	0.11	
					Winter PDO $\rightarrow$ June (E GOA) upwelling	0.06	-0.37		
					Winter PDO $\rightarrow$ Fall (Total) discharge	0.07	-0.30	0.09	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.01	-0.54	NA	
All	0.72	0.11	30.39	13	Winter/Spring (W GOA) Upwelling $\rightarrow$ Recruitment	0.27	-0.19	0.05	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.87	-0.03		
					NPGO $\rightarrow$ Recruitment	0.62	0.09		
					Winter NOI $\rightarrow$ Winter/Spring (W GOA) upwelling	0.03	0.38	0.28	
					Winter PDO $\rightarrow$ Winter/Spring (W GOA) upwelling	0.23	-0.21		
					Winter PDO $\rightarrow$ Fall (Total) Discharge	0.07	-0.30	0.09	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.01	-0.54	NA	
All	0.71	0.09	31.03	13	June (W GOA) Upwelling $\rightarrow$ Recruitment	0.00	-0.64	0.42	Failure to reject
					Fall (Total) Discharge $\rightarrow$ Recruitment	0.52	-0.09		
					NPGO $\rightarrow$ Recruitment	0.31	-0.14		
					Winter NOI $\rightarrow$ June (W GOA) upwelling	0.25	-0.23	0.05	
					Winter PDO $\rightarrow$ June (W GOA) upwelling	0.28	-0.22		
					Winter PDO $\rightarrow$ Fall (Total) discharge	0.07	-0.30	0.09	
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.01	-0.54	NA	
H_{A1} & H_{A3}	0.64	0.03	21.17	10					Failure to reject
H_{A1} & H_{A3}	0.60	0.02	22.69	10					Failure to reject
H_{A1} & H_{A3}	0.68	0.06	19.57	10	June (E GOA) Upwelling $\rightarrow$ Recruitment	0.05	-0.32	0.11	Failure to reject
					NPGO $\rightarrow$ Recruitment	0.85	-0.03		
					Winter NOI $\rightarrow$ June (E GOA) upwelling	0.14	-0.29	0.11	
					Winter PDO $\rightarrow$ June (E GOA) upwelling	0.06	-0.37		
					Winter PDO $\leftarrow \rightarrow$ Winter NOI	0.01	-0.55	NA	
H_{A1} & H_{A3}	0.64	0.03	21.42	10					Failure to reject
H_{A1} & H_{A3}	0.61	0.02	22.41	10					Failure to reject
H_{A2} & H_{A3}	0.50	0.00	17.87	7					Failure to reject
H_{A2} & H_{A3}	0.75	0.12	9.38	7	Fall (Total) Discharge $\rightarrow$ Recruitment	0.72	-0.06	0.00	Failure to reject
					NPGO $\rightarrow$ Recruitment	0.88	0.03		
					Winter PDO $\rightarrow$ Fall (Total) discharge	0.07	-0.30	0.09	
H_{A1} & H_{A2}	0.48	0.00	26.52	11					Failure to reject

Table 3 (continued)

Hypotheses tested	Adjusted goodness-of-fit	Pr ($> \chi^2$)	$\Delta AICc$	K	Path	Path p-value	Path β -estimate	Endogenous variable R^2	H ₀ outcome
H _{A1} & H _{A2}	0.49	0.00	26.08	11					Failure to reject
H _{A1} & H _{A2}	0.50	0.01	25.99	11					Failure to reject
H _{A1} & H _{A2}	0.52	0.01	24.74	11					Failure to reject
H _{A1} & H _{A2}	0.49	0.00	26.59	11					Failure to reject
H _{A1} & H _{A2}	0.69	0.09	19.34	11	April (E GOA) Upwelling → Recruitment	0.54	−0.11	0.02	Failure to reject
					Fall (Total) Discharge → Recruitment	0.66	−0.08		
					Winter NOI → April (E GOA) upwelling	0.76	0.06	0.01	
					Winter PDO → April (E GOA) upwelling	0.87	−0.03		
					Winter PDO → Fall (Total) discharge	0.07	−0.30	0.09	
					Winter PDO ← → Winter NOI	0.01	−0.54	NA	
H _{A1} & H _{A2}	0.67	0.07	19.90	11	May (E GOA) Upwelling → Recruitment	0.98	0.00	0.00	Failure to reject
					Fall (Total) Discharge → Recruitment	0.73	−0.06		
					Winter NOI → May (E GOA) upwelling	0.10	−0.33	0.10	
					Winter PDO → May (E GOA) upwelling	0.88	−0.03		
					Winter PDO → Fall (Total) discharge	0.07	−0.30	0.09	
					Winter PDO ← → Winter NOI	0.01	−0.54	NA	
H _{A1} & H _{A2}	0.76	0.20	17.23	11	June (E GOA) Upwelling → Recruitment	0.05	−0.33	0.12	Failure to reject
					Fall (Total) Discharge → Recruitment	0.56	−0.10		
					Winter NOI → June (E GOA) upwelling	0.14	−0.29	0.11	
					Winter PDO → June (E GOA) upwelling	0.06	−0.37		
					Winter PDO → Fall (Total) discharge	0.07	−0.30	0.09	
					Winter PDO ← → Winter NOI	0.01	−0.54	NA	
H _{A1} & H _{A2}	0.73	0.14	18.21	11	Winter/Spring (W GOA) Upwelling → Recruitment	0.34	−0.16	0.03	Failure to reject
					Fall (Total) Discharge → Recruitment	0.86	−0.03		
					Winter NOI → Winter/Spring (W GOA) upwelling	0.03	0.38	0.28	
					Winter PDO → Winter/Spring (W GOA) upwelling	0.23	−0.21		
					Winter PDO → Fall (Total) discharge	0.07	−0.30	0.09	
					Winter PDO ← → Winter NOI	0.01	−0.54	NA	
H _{A1} & H _{A2}	0.74	0.15	17.89	11	June (W GOA) Upwelling → Recruitment	0.00	−0.60	0.37	Failure to reject
					Fall (Total) Discharge → Recruitment	0.51	−0.09		
					Winter NOI → June (W GOA) upwelling	0.25	−0.23	0.05	
					Winter PDO → June (W GOA) upwelling	0.28	−0.22		
					Winter PDO → Fall (Total) discharge	0.07	−0.31	0.09	
					Winter PDO ← → Winter NOI	0.01	−0.54	NA	
0.55	0.03	12.99	12.19	8					Failure to reject
0.55	0.03	13.03	12.23	8					Failure to reject
H _{A1}	0.66	0.07	10.21	8	June (E GOA) Upwelling → Recruitment	0.05	−0.32	0.10	Failure to reject
					Winter NOI → June (E GOA) upwelling	0.14	−0.29	0.11	
					Winter PDO → June (E GOA) upwelling	0.06	−0.37		
					Winter PDO ← → Winter NOI	0.01	−0.54	NA	
H _{A1}	0.58	0.03	11.69	8					Failure to reject
H _{A1}	0.57	0.03	11.76	8					Failure to reject
H _{A2}	−0.04	0.00	13.75	5					Failure to reject
H _{A2}	0.48	0.02	6.39	5					Failure to reject
H _{A3}	1.00	NA	0.00	3	NPGO → Recruitment	0.89	0.02	NA	Failure to reject

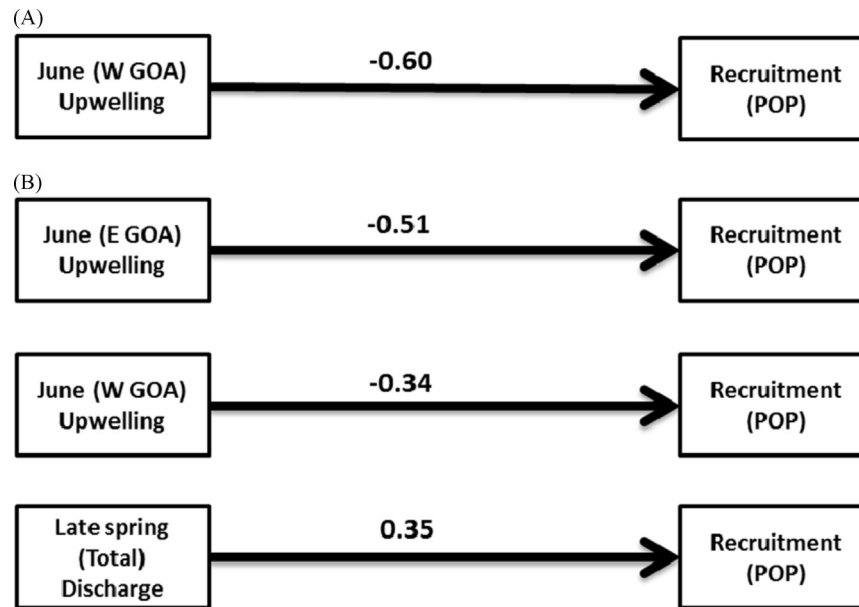


Fig. 8. Path diagrams mapping significant paths to POP recruitment found in candidate models from exploratory phase of SEM analysis. Only significant paths between environmental variables and recruitment are represented, including significant sequential paths. Diagrams are shown for models testing relationships during age 0 (a) and age 1 (b); no significant paths to recruitment were found during age 2. Arrows point from exogenous to endogenous variables. Numbers denote parameter estimates.

the year are associated with longer periods of weak water column stability accompanied by intermittent vertical mixing, longer phytoplankton blooms, and greater zooplankton productivity (Eslinger et al., 2001; Henson, 2007). This may enhance availability of the copepods and euphausiids that comprise the largest portion of juvenile POP diets (Carlson and Haight, 1976; Boldt and Rooper, 2009).

4.3. Implications and limitations

This study lends further insight into the idea that the shelves of the E and W GOA are subject to very different environmental influences. Downwelling patterns show a stark contrast between the E and W GOA, with the latter showing greater upwelling-favorable winds throughout much of the year relative to the E GOA, particularly during summer and fall. In contrast, the E GOA exhibits a clearer seasonal pattern than the W GOA, with downwelling-favorable winds throughout most of the year and a period of relaxed downwelling and upwelling during summer (Fig. 4). In the light of these differences in variability, it is noteworthy that upwelling index variability in both regions of the GOA (during late spring) were related to POP recruitment, whereas only variability in the E GOA (during summer) was related to sablefish recruitment. This may reflect the higher density of young sablefish in the E GOA (Heifetz and Fujioka, 1991; Sigler et al., 2001; Maloney and Sigler, 2008), whereas juvenile POP are widely distributed throughout the northern GOA and Aleutian Islands (Rooper, 2008).

One of the concerns with this analysis is the use of SEM with relatively short time series that may have non-linear relationships. While SEM is a promising approach to testing relationships among large-scale climate drivers, regional environmental variability, and local or regional biological responses, several factors may limit the utility of SEM in this study. First, SEM is based on linear regression and requires appropriate transformations, for example exponential terms, to evaluate nonlinear relationships (Bollen, 1989; Ullman et al., 2007). Since only linear relationships were hypothesized, it is possible that significant nonlinear relationships were overlooked. Second, SEM is a large-sample technique. Kline (1998) recommended a minimum sample size of 10 times the number of parameters, which our larger models failed to meet, and all of our models failed to meet the sample sizes recommended by MacCallum et al. (1996).

Additionally, while we took precautions to reduce the risk of spurious relationships, the large number of structural models tested involves a high risk of type I error. It is common for environment–recruit relationships to appear robust when first analyzed, only to break down upon later investigation with updated data (Myers, 1998). Due to the increased risk of spurious findings in our study, it will be particularly important to reassess the reliability of our results in the future.

Another complication for interpreting our results is the population structure of POP, which consists of numerous genetically distinct subpopulations throughout the GOA (Palof et al., 2011; Kamin et al., 2013). The authors hypothesize that this is not due to a sweepstakes effect, but due to limited physical dispersal at all life stages. Thus, a gulfwide recruitment index for POP in the GOA may mask differences in local recruitment patterns and links between the gulfwide recruitment index and our environmental indices may be weak even if upwelling and discharge have local effects on some of the GOA POP subpopulations. While geographically isolated subpopulations are not hypothesized for sablefish, relationships between recruitment and regional environmental conditions may still be difficult to identify based on statewide recruitment indices. Thus, for both species, regional indices of recruit abundance may better serve to detect relationships between regional environmental variables and recruitment.

It is also worth noting that juvenile sablefish and POP may have swimming capabilities that make them less susceptible to advection than we assume in our study. Results of some studies indicate that other *Sebastes* species may be able to swim against currents during their pelagic stage (e.g., Nishimoto and Washburn, 2002; Miller and Shanks, 2004; Kashaf et al., 2014). Sablefish develop deeply forked caudal tails early in their life history, enabling faster swimming speeds (Shaw and McFarlane, 1997); furthermore, some suggest that their high growth rate may increase their swimming ability throughout their early life history (e.g., Shenker and Olla, 1986). Our assumption that these species are primarily passive drifters during their juvenile pelagic stage has not been verified.

Finally, recruitment is a complex process susceptible to factors operating at multiple life stages (e.g., Kruse and Tyler, 1989; Wilderbuer et al., 2002). Here we focused on factors associated with larval drift and found no support for our *a priori* hypotheses, but exploratory analyses revealed relationships that may imply

productivity-related mechanisms and may act during later life stages. When more information on factors such as the locations of hatching and parturition becomes available, examining conditions prior to their larval drift period may also be informative.

Our results point to potential mechanisms warranting further investigation using more targeted process studies. We identified variables that may indicate advection and productivity-related mechanisms affecting recruitment of sablefish and POP, as well as the time periods during which these mechanisms may operate. Environment–recruit relationships are sometimes incorporated into simulated population projections performed during MSEs in order to evaluate the performance of different candidate management strategies under alternate scenarios (Smith, 1994; A'Mar et al., 2009; Ianelli et al., 2011). However, incorporating statistically significant environmental indices into these projections often fails to result in any substantial difference in management strategy performance (Punt, 2011; Punt et al., 2014). Environment–recruit relationships tend to be more useful to MSEs when the underlying mechanisms are well understood and when the environmental variable can be reliably predicted (Basson, 1999; Punt et al., 2014). Using SEM allowed us to incorporate mechanistic assumptions into our statistical models; however, before the usefulness of incorporating these results into future MSEs can be determined, process studies would need to be conducted to confirm the proposed mechanisms.

While the mechanisms behind the relationships found in our study are uncertain, our methods and results may still be useful to researchers. It would be beneficial to revisit this analysis in future years when longer time series are available for the variables used (e.g., Myers, 1998). In the meantime, our results suggest potentially informative relationships that are worth further investigation, and which may provide valuable insight into the recruitment dynamics of these two species regardless of whether or not this information turns out to be useful in MSEs. It may be beneficial to analyze these relationships individually using approaches that relax the linearity assumption, such as generalized additive models, until sample sizes are large enough to more reliably incorporate sequential causation into model structure using SEM.

Acknowledgments

The authors wish to thank Professor Gordon Kruse and Dr. S. Kalei Shotwell for the substantial roles each played in guiding the analysis and writing for this study. Dr. Jason Waite, Dr. Dana Hanselman, Dr. Miriam Doyle, and Danielle Dickson also contributed valuable information, suggestions, and data. We also wish to thank three anonymous reviewers for their feedback and advice. This study was funded by the North Pacific Research Board (NPRB, Grant G81, publication #527) as part of the Gulf of Alaska Integrated Ecosystem Research Project (GOA-IERP, publication #2).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.dsr2.2015.02.016>.

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Update

Deep-Sea Research Part II

Volume 134, Issue , December 2016, Page 426

DOI: <https://doi.org/10.1016/j.dsr2.2016.10.002>



Corrigendum to “Environmental covariates of sablefish (*Anoplopoma fimbria*) and Pacific ocean perch (*Sebastes alutus*) recruitment in the Gulf of Alaska” [Deep-Sea Res. II 132 (2016) 194–209]



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Figure 4 on page 200 erroneously shows downwelling index values by Julian day and longitude for 2011 only, rather than long-term averages for 1967–2011. The corrected figure (Fig. 1) shows very similar but smoother patterns overall, including more obvious differ-

ences between the eastern and western Gulf of Alaska. The discrepancies between the original and corrected figures do not affect the overall interpretation of results.

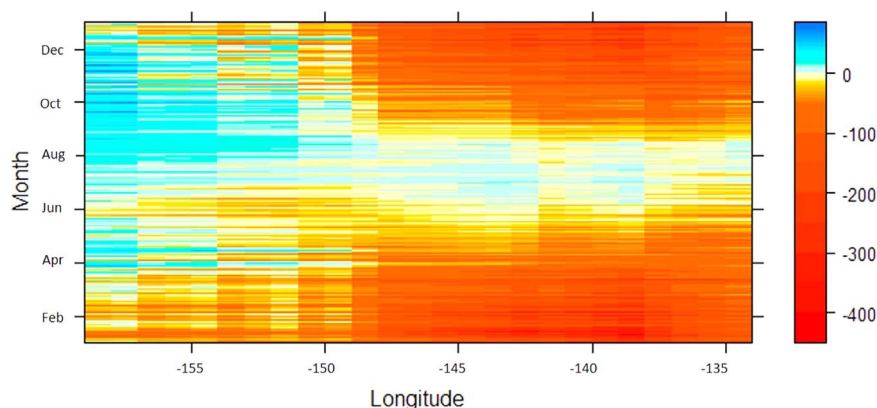


Fig. 1. Hovmöller diagram showing downwelling index values by month and longitude. Daily values are averaged across years 1967–2011 and longitude is on a 1° resolution from west to east. Warm colors (negative values) represent downwelling-favorable winds and cool colors (positive values) represent upwelling-favorable winds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

DOI of original article: <http://dx.doi.org/10.1016/j.dsr2.2015.02.016>

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<http://dx.doi.org/10.1016/j.dsr2.2016.10.002>

Available online 15 October 2016
0967-0645/